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Domain Generalized Heterogeneous Graph Neural Networks for Cross-Domain Financial Fraud Detection

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ABSTRACT Financial transactions are used widely across multiple digital platforms. It makes them essential to the modern economy. However, due to this rapid growth there has also been an increase in the scale and complexity of financial fraud. Evolving fraud patterns make it difficult for traditional fraud detection models to capture complex relationships between financial entities and transactions. In this paper, we implement a Domain Generalized Heterogeneous Graph Neural Network (DG-Hetero-GNN) framework for finding robust financial fraud detection under distribution shifts. This proposed framework applies to domain generalization in order to improve performance across unseen domains through Leave-One-Domain-Out (LODO), Out-of-Distribution (OOD), and adaptive training settings. Experiments conducted on multiple financial transaction datasets demonstrate that the proposed framework achieves a strong cross-domain generalization while maintaining high fraud detection capability under varying transaction distributions.

INDEX TERMS Domain Generalization , DG-Hetero-GNN , Cross-Domain , LODO ,OOD

I. INTRODUCTION

Financial transactions have become a fundamental component of the modern economy spanning across applications such as banking, creditcard systems, mobile payments and cryptocurrency platforms. With the advancement of these services, the volume and complexity of transactions increased significantly. This leads to a rise in fraudulent activities. Fraudulent activities are challenging due to their imbalanced datasets, evolving fraud patterns and complex relationships between entities such as users, transactions and devices [8], [12]. This challenge makes traditional machine learning approaches not feasible as they rely on tabular data. Because of this, it makes it hard to capture relational dependencies.

Then Graph Neural Networks (GNNs) emerged as an effective approach for modelling structured data by leveraging entities called as nodes [7], [10]. Through the propagation of information connected

across nodes, GNNs can capture hidden patterns in transaction networks. This enables GNNs to detect fraud behaviours coordinated with each other. Several variants of this model have been proposed, including attention-based models that dynamically assign importance to neighboring nodes [20] as well as their architectures that improve representation power for graph learning tasks [19]. These advancements have improved performance in graph-based applications.

In order to represent real-world systems, studies [13] have shown the multiple node and edge types within heterogeneous financial graphs. Heterogeneous graph learning approaches improve representation learning by capturing complex interactions among transactions, accounts, merchants, and devices within financial transaction networks.

Despite these advancements, most existing GNN-based fraud detection models assume that the training and testing data follow similar distributions. In

real world applications fraud detection systems are deployed across multiple domains, where transaction patterns, feature distributions, and graph structures differ. This distribution shift leads to poor generalization when models are applied to unseen datasets. Domain Generalization techniques aim to address this issue by learning invariant representations across [6], [11]. However, recent studies have shown that many existing methods struggle to maintain consistent performance in practical settings [12], while domain adaptation approaches often require access to target domain data, limiting their applicability [14].

Another challenge of fraud detection is the extreme class imbalance, where fraudulent transactions constitute only a small fraction of the dataset. Due to this imbalance, metrics such as PR-AUC, ROC-AUC, Precision, Recall, and F1-score are widely used to evaluate model performance [2], [15]. These metrics have proven to be a better assessment of fraud detection capability under highly imbalanced financial environments. However, in fraud detection it is usually preferred to use metrics supportive to F1 score to address its stability. In addition to this, specialized learning strategies are necessary to address imbalanced data distributions [8]. Representation of learning approaches, including random-walk-base methods, have also been explored to capture structural data in graphs [17].

Recent research has explored the integration of heterogeneous graph neural networks (HGNNs) with cross domain learning strategies for fraud detection [12], [26]. More advanced approaches focus on improving robustness and adaptability in dynamic and evolving environments [4], [24]. However, achieving consistent and reliable performance across multiple domains remains a challenge.

To address these limitations, we built DG-Hetero-GNN, a Domain Generalized Heterogeneous Graph Neural Network framework for cross-domain fraud detection. This framework combines heterogeneous GraphSAGE message passing, adaptive domain generalization, focal loss optimization, and adaptive threshold selection to learn domain-invariant representations across multiple financial transaction datasets. Unlike existing approaches, DG-Hetero-GNN improves cross-domain generalization while maintaining stable fraud detection performance under different feature distributions and heterogeneous graph structures.

Furthermore, we evaluate the model on multiple datasets under:

- Leave-One-Domain-Out (LODO)
- Out-Of-Domain (OOD)
- Adaptive OOD

II. RELATED WORK

A. GRAPH NEURAL NETWORKS

Traditional fraud detection methods rely on tabular machine learning models which treat transactions as independent instances [1], [8]. Such types of approaches fail on capturing relational dependencies between entities. These relational dependencies are critical for identifying coordinated and complex patterns. Additionally, these models are highly sensitive to class imbalance and struggle to generalize across datasets with variable distributions. This leads to a reduction in performance in real-world applications [6], [7], [14].

Then came Graph Neural Networks (GNNs), where nodes represent entities and edges capture interactions between them. By utilizing neighborhood aggregations, GNNs can learn representations that incorporate both feature and structural information. Foundational models such as Graph Convolution Networks (GCNs) [10], GraphSAGE [7] and Graph Attention Networks (GATs) [20] have shown an improved performance in fraud detection by capturing hidden patterns in transaction networks. Furthermore, studies have proven that GNNs show strong representation capabilities for graph-structured data. This helps them to enable complex relationships effectively than traditional methods [19].

B. HETEROGENEOUS GRAPH NEURAL NETWORKS

GNN-based approaches assume homogeneous graph structures and consistent data distributions between training and testing environments. In real world transaction systems, interactions involve multiple types of entities, and data distributions differ across domains. As a result, homogeneous GNNs often fail to capture complexity of financial networks and show limited generalization capability under distribution shifts.

Heterogeneous Graph Neural Networks (HGNNs) have been proposed to model multiple types of nodes and relationships. These models extend traditional frameworks by using diverse entity transactions. This enables them for more expressive and semantically rich representations. Approaches such

as heterogeneous graph learning methods [13] and relational graph architectures [19] improve representation learning by modeling multiple node and edge types within financial transaction networks. Recent studies have shown that application of heterogeneous graphs learning, specifically for fraud detections in multi-relational dependencies, tend to be effective in uncovering coordinated fraudulent behaviors [12], [13].

Despite these advantages, HGNNs are evaluated in single-domain settings and do not address cross-domain generalization. In practical scenarios, fraud detection systems must operate across datasets with varying feature distributions and graph structures. Under such conditions, HGNNs often learn domain specific patterns. This leads to performance degradation when applied to unseen domains. Our model is designed to improve robustness and maintain a stable performance across multiple financial datasets.

III. PROBLEM DEFINITION

Heterogeneous Graph A heterogeneous graph is a directed graph $G = (V, E, A, R)$, where V denotes the set of nodes, E represents the set of edges, A shows the set of node type and R depicts the set of edge types. Each node $v \in V$ and $e \in E$ are associated with type mapping functions $\tau(V) : V \rightarrow A$ and $\phi(e) : E \rightarrow R$. This formulation helps modeling of multiple types of entities and relationships in financial systems such as transactions, users and merchants.

Graph Based Fraud Detection: Given the heterogeneous graph G with node feature matrix X and node set V , the fraud detection task is put together as a node classification problem. The objective is to learn a predictive function f_θ such that:

$$Y = f_\theta(X, G) \quad (1)$$

where $Y \in \{0, 1\}$, with 1 showing fraud nodes and 0 showing legitimate nodes. This model is trained in a semi-supervised manner using partially labeled nodes.

Cross-Domain Setting: In real world applications, fraud detection systems run across multiple domains with different feature distributions and graph structures. Let $\mathcal{G} = \{G^1, G^2, \dots, G^D\}$ indicate a set of heterogeneous graphs from different domains. The model is trained on a subset of source domains $\mathcal{G}_s \subset \mathcal{G}$ and evaluated on an unseen target domain $G^t \notin \mathcal{G}_s$.

IV. OUR MODEL

A. OVERVIEW

In general, Domain Generalized Heterogeneous Graph Neural Network (DG-Hetero-GNN) follows an aggregation based graph learning framework defined as:

$$H[t] \leftarrow \text{Aggregate}_{\forall s \in \mathcal{N}(t), \forall r \in R} \text{Message}(s, r, t) \quad (2)$$

where t denotes the target node, $\mathcal{N}(t)$ represents the set of neighboring nodes, and R denotes the set of edge types. Each neighboring node $s \in \mathcal{N}(t)$ may belong to different node and relation types across multiple domains. This formulation follows the standard message passing mechanism widely used in graph neural networks [7], [10], [19].

This framework is comprised of three main components:

- **Message:** extracts information from the source node and relation;
- **Aggregate:** combines information from all neighboring nodes;
- **Update:** updates the representation of the target node.

As shown in Fig. 1, the proposed DG-Hetero-GNN framework consists of the following main modules:

- (1) Type-Specific Feature Projection
- (2) Heterogeneous GraphSAGE Message Passing
- (3) Fraud Classification Module
- (4) Adaptive Domain Generalization
- (5) Threshold Optimization

B. TYPE-SPECIFIC FEATURE PROJECTION

Since heterogeneous node types may contain different feature dimensions, type-specific feature encoders are used to align heterogeneous node features into a shared latent representation space.

$$H_{proj}^{(k)} = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(W_k X^{(k)}))) \quad (3)$$

where $X^{(k)}$ represents the feature matrix of node type k , and W_k denotes a learnable projection matrix. This projection step enables consistent representation learning across heterogeneous financial entities and domains.

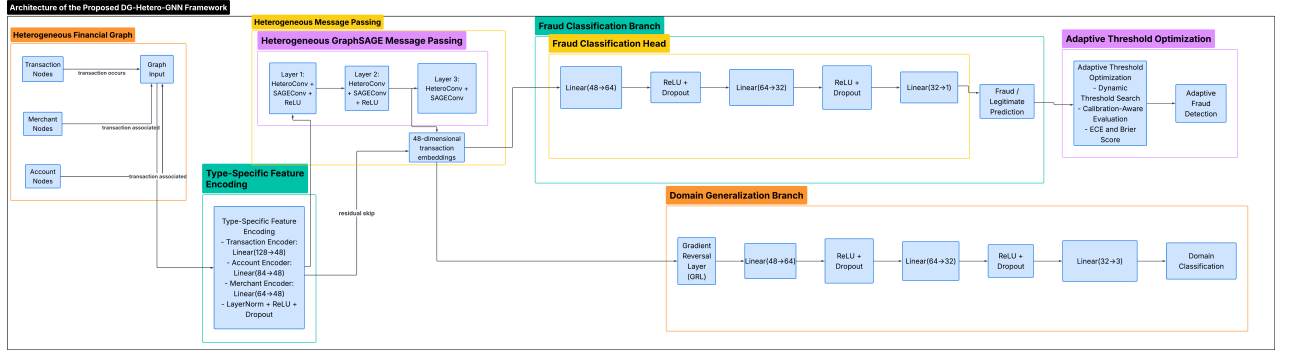


FIGURE 1: Architecture of the proposed DG-Hetero-GNN model

C. HETEROGENEOUS GRAPHSAGE MESSAGE PASSING

In order to capture relational dependencies between financial entities, heterogeneous message passing is performed across multiple edge types.

$$H^{(l+1)} = \text{HeteroConv} \left(\{ \text{SAGEConv}_r(H_u^{(l)}, H_v^{(l)}) \}_{r \in R} \right) \quad (4)$$

where relation-specific GraphSAGE convolution operators are applied for each edge type $r \in R$.

Three stacked heterogeneous GraphSAGE layers are used to aggregate neighborhood information and capture multi-relational interactions between entities across domains.

Nonlinear activation functions are applied after each message-passing layer to refine node representations and improve representation learning stability.

D. FRAUD CLASSIFICATION MODULE

The final transaction node embeddings are passed through a multi-layer perceptron classifier for binary fraud prediction:

$$\hat{Y} = \sigma(WH + b) \quad (5)$$

here σ denotes the sigmoid activation function, W represents the learnable weight matrix, and b denotes the bias term. The classifier produces fraud probability scores for transaction nodes.

E. ADAPTIVE DOMAIN GENERALIZATION

To improve robustness across unseen domains, the implemented framework uses a Gradient Reversal Layer (GRL) together with a domain classifier. The GRL encourages the model to learn domain-invariant node representations by maximizing domain confusion while preserving fraud classification performance.

The overall optimization objective is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{fraud} + \lambda \mathcal{L}_{domain} \quad (6)$$

where $\mathcal{L}_{fraud}$ denotes the fraud classification loss, $\mathcal{L}_{domain}$ represents the domain classification loss, and λ controls the contribution of domain adaptation during training.

F. FOCAL LOSS OPTIMIZATION

Financial fraud detection datasets are highly imbalanced due to their relatively small number of fraudulent transactions. To address this issue, the proposed framework is optimized using focal loss:

$$\mathcal{L}_{focal} = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (7)$$

where p_t denotes the predicted fraud probability, α controls class weighting, and γ adjusts the focusing parameter for difficult samples. This loss function improves fraud sensitivity and reduces the impact of class imbalance during training.

G. THRESHOLD OPTIMIZATION

In order to improve fraud sensitivity under distribution shifts, optimal decision thresholds are determined using a validation-based F1-score maximization.

$$T^* = \arg \max_T F1(T) \quad (8)$$

where T^* represents the optimal classification threshold selected to maximize the F1-score. This adaptive threshold optimization improves robustness across unseen financial domains.

H. ALGORITHM

The overall training procedure of the model is summarized in Algorithm IV-H. The algorithm outlines the type-specific feature projection, heterogeneous

GraphSAGE message passing, fraud classification, adaptive domain generalization, and threshold optimization procedures across multiple domains.

Require: Heterogeneous graph set $\mathcal{G}_s$, node features X , labels Y

Ensure: Trained model parameters θ

```

1: Initialize model parameters  $\theta$ 
2: for each epoch do
3:   for each graph  $G \in \mathcal{G}_s$  do
4:     Perform type-specific feature projection using (3)
5:     Perform heterogeneous GraphSAGE message passing using (4)
6:     Generate transaction node embeddings
7:     Compute fraud predictions using (5)
8:     Compute focal loss using (7)
9:     if adaptive training enabled then
10:      Apply Gradient Reversal Layer (GRL)
11:      Compute domain classification loss using (6)
12:      Combine fraud and domain losses
13:     end if
14:     Update model parameters  $\theta$  using back-propagation
15:   end for
16: end for
17: Perform threshold optimization using (8)
18: return  $\theta$ 

```

V. EXPERIMENTS

In this experimental section, we mainly present:

- Heterogeneous graph construction from real world fraud data;
- Performance comparison with baseline models under cross-domain settings;
- Evaluation under Leave-One-Domain-Out (LODO), Out-of-Domain (OOD), and adaptive cases;
- Analysis of model generalization and robustness across different domains.

A. EXPERIMENTAL SETUP

1) Dataset

We use Elliptic [22], DGraphFin [9], AMLSim [23], and IEEE-CIS [21]. These datasets show different financial domains with different feature distributions, graph structures, and scales.

The **Elliptic** dataset consists of transaction-level features obtain from temporal and structural properties of bitcoin network. Each transaction node is represented by handcrafted features capturing transaction

behaviors and their graph connectivity patterns.

The **DGraphFin** dataset is a large-scale financial graph dataset containing transaction nodes and relational interactions for graph anomaly detection. The dataset includes heterogeneous financial entities with structural and temporal transaction relationships.

The **AMLSim** dataset is a synthetic anti-money laundering transaction dataset used to simulate realistic financial transaction behaviors and illicit money laundering activities across multiple entities and transaction patterns.

The **IEEE-CIS** dataset includes high-dimensional transaction features which capture detailed behavioral, temporal, and identity based information, allowing richer representation of fraud patterns.

To model complex relationships, each dataset is turned into a heterogeneous graph based on domain knowledge.

Transaction records are shown as transaction (Txn) nodes, while associated entities such as accounts (Account) and merchants (Merchant) are shaped as separate node types.

Table 1 summarizes the nodes statistics of the constructed heterogeneous graphs. We can see that different datasets show variation in node types, feature dimensions and scale. In particular, the contrast between high dimensional IEEE features and the structural diverse DGraphFin and AMLSim graphs introduce domain shifts making cross-domain fraud detection a challenge.

2) Baselines

The performance of our model DG-Hetero-GNN is verified by comparing it with classical machine learning models, deep learning approaches, and graph neural network baselines. For traditional methods, we use XGBoost [1]. For deep learning models, we include a Multi-Layer Perceptron (MLP) [5]. For graph neural network baselines, we use Graph Convolutional Networks (GCN) [10], GraphSAGE [7], and Graph Attention Networks (GAT) [20]. In addition, we also implement Heterogeneous GraphSAGE (HeteroGraphSAGE). This is an extension of GraphSAGE aggregation [7] as a heterogeneous graph neural network baseline. XGBoost and MLP operate on feature-based representations without directly modeling graph structure, while GCN, GraphSAGE, and GAT are designed for homogeneous graph learning through neighborhood aggregation. Heterogeneous GraphSAGE extends GraphSAGE aggregation to heterogeneous financial graphs by capturing interactions across multiple

TABLE 1: Statistics of Constructed Heterogeneous Financial Graphs

Dataset	Transaction Nodes	Entity Nodes	Feature Size	Total Edges
Elliptic	46,564	93,128	128	222,880
IEEE-CIS	590,540	14,962	128	2,952,700
DGraphFin	250,000	324,822	128	1,036,164
AMLSim	45	35	128	443

TABLE 2: LODO Cross-Domain Evaluation Results

Model	ROC-AUC			PR-AUC			F1-Score		
	LODO ₀	LODO ₁	LODO ₂	LODO ₀	LODO ₁	LODO ₂	LODO ₀	LODO ₁	LODO ₂
DG-Hetero-GNN	0.7126 ± 0.0445	0.5879 ± 0.0226	0.6066 ± 0.0202	0.1819 ± 0.0462	0.0451 ± 0.0019	0.0854 ± 0.0044	0.2449 ± 0.0252	0.0885 ± 0.0058	0.1514 ± 0.0069
GAT	0.713 ± 0.03	0.59 ± 0.02	0.61 ± 0.02	0.17 ± 0.03	0.05 ± 0.01	0.08 ± 0.01	0.22 ± 0.02	0.09 ± 0.01	0.15 ± 0.01
GCN	0.64 ± 0.03	0.57 ± 0.02	0.58 ± 0.02	0.11 ± 0.02	0.04 ± 0.01	0.07 ± 0.01	0.17 ± 0.02	0.08 ± 0.01	0.13 ± 0.01
GraphSAGE	0.62 ± 0.02	0.55 ± 0.02	0.57 ± 0.02	0.10 ± 0.01	0.04 ± 0.01	0.07 ± 0.01	0.15 ± 0.01	0.08 ± 0.01	0.13 ± 0.01
HeteroGraphSAGE	0.60 ± 0.04	0.39 ± 0.05	0.48 ± 0.04	0.08 ± 0.01	0.03 ± 0.01	0.05 ± 0.01	0.12 ± 0.01	0.05 ± 0.01	0.08 ± 0.01
MLP	0.62 ± 0.06	0.55 ± 0.03	0.56 ± 0.03	0.09 ± 0.02	0.04 ± 0.01	0.06 ± 0.01	0.14 ± 0.02	0.07 ± 0.01	0.12 ± 0.01
XGBoost	0.5293 ± 0.0983	0.5946 ± 0.0502	0.6078 ± 0.0131	0.1034 ± 0.0235	0.0554 ± 0.0094	0.0870 ± 0.0051	0.2375 ± 0.0351	0.0349 ± 0.0196	0.1368 ± 0.0666

TABLE 3: Detailed LODO Analysis of DG-Hetero-GNN

Dataset	Precision	Recall	Threshold	ECE ↓	Brier ↓	TP	TN	FP
LODO ₀	0.1449 ± 0.0236	0.8655 ± 0.1314	0.3990 ± 0.0020	0.2999 ± 0.0098	0.1760 ± 0.0057	3933.6000 ± 597.0898	17626.6000 ± 6820.3672	24392.4000 ± 6820.3672
LODO ₁	0.0495 ± 0.0036	0.5095 ± 0.1993	0.1700 ± 0.0773	0.1194 ± 0.0179	0.0617 ± 0.0042	10527.2000 ± 4118.8285	363248.8000 ± 89917.8804	206628.2000 ± 89917.8804
LODO ₂	0.0887 ± 0.0041	0.5189 ± 0.0429	0.3850 ± 0.0176	0.2872 ± 0.0146	0.1432 ± 0.0095	8047.4000 ± 665.3085	151762.8000 ± 6886.7996	82728.2000 ± 6886.7996

TABLE 4: OOD AMLSim Evaluation Results

Model	ROC-AUC	PR-AUC	F1-Score	ECE ↓	Brier ↓
DG-Hetero-GNN	0.5852 ± 0.0502	0.6290 ± 0.0519	0.7243 ± 0.0078	0.2638 ± 0.0186	0.3033 ± 0.0078
GAT	0.56 ± 0.04	0.61 ± 0.04	0.70 ± 0.02	0.21 ± 0.02	0.27 ± 0.02
GCN	0.53 ± 0.04	0.58 ± 0.04	0.66 ± 0.02	0.25 ± 0.02	0.29 ± 0.02
GraphSAGE	0.52 ± 0.03	0.56 ± 0.03	0.64 ± 0.02	0.22 ± 0.02	0.26 ± 0.02
HeteroGraphSAGE	0.58 ± 0.05	0.59 ± 0.05	0.68 ± 0.03	0.31 ± 0.03	0.33 ± 0.03
MLP	0.50 ± 0.05	0.55 ± 0.05	0.63 ± 0.03	0.20 ± 0.02	0.24 ± 0.02
XGBoost	0.5616 ± 0.0177	0.6330 ± 0.0352	0.0154 ± 0.0308	0.0274 ± 0.0016	0.0344 ± 0.0002

node and edge types. Compared to these baselines, our model DG-Hetero-GNN is designed to efficiently handle cross-domain distribution shifts while leveraging heterogeneous graph structures, adaptive threshold optimization, and domain generalization for improved fraud detection.

3) Experimental Setting

To evaluate cross domain generalization of our

model, we adopt multiple experimental settings such as Leave-One-Domain-Out (LODO), Out-of-Domain (OOD) and adaptive training cases. In the LODO setting, the model is trained on all but one dataset and evaluated on the held-out domain. This process is repeated across all datasets to test generalization under partial domain shifts. Here we define LODO₀, LODO₁, LODO₂. LODO₀ is trained on IEEE and DGraphFin datasets and tested on Elliptic.

TABLE 5: Adaptive OOD Evaluation of DG-Hetero-GNN

Metric	OOD	Adaptive
ROC	0.5852 $\pm$ 0.0502	0.5848 $\pm$ 0.0465
PR	0.6290 $\pm$ 0.0519	0.6319$\pm$0.0477
Prec	0.6187 $\pm$ 0.0502	0.6230$\pm$0.0465
Recall	0.8960$\pm$0.1120	0.8800 $\pm$ 0.1012
F1	0.7243$\pm$0.0078	0.7232 $\pm$ 0.0082
Thr	0.1750 $\pm$ 0.0550	0.1930 $\pm$ 0.0638
ECE $\downarrow$	0.2638 $\pm$ 0.0186	0.2628$\pm$0.0209
Brier $\downarrow$	0.3033 $\pm$ 0.0078	0.3019$\pm$0.0095
TP	22.4 $\pm$ 2.8	22.0 $\pm$ 2.5
TN	5.6 $\pm$ 4.5	6.2$\pm$4.1
FP	14.4 $\pm$ 4.5	13.8$\pm$4.1
FN	2.6$\pm$2.8	3.0 $\pm$ 2.5

LODO1 is trained on Elliptic and DGraphFin and tested on IEEE. LODO2 is trained on Elliptic and IEEE and tested on DGraphFin.

Similarly, in the OOD setting, it trains on multiple source domains and is tested on an unseen target domain representing a realistic and challenging scenario. In our paper we use AMLSim as the unseen target domain dataset. In Table 1 you can see that the nodes and its types are less. The AMLSim dataset is reduced for preserving diversity as it is highly imbalanced. This makes OOD meaningful and efficient for testing.

In the adaptive setting the model is trained using both the source and the target domain data to analyze and assess the impact of domain adaptation.

All the models are implemented using PyTorch and PyTorch Geometric. For fair comparison, consistent hyperparameter settings are used across all methods. The input node features are standardized to 128 dimensions for transaction nodes and 64 dimensions for account and merchant nodes, while the hidden embedding dimension is fixed at 48. A three-layer heterogeneous GraphSAGE architecture is implemented for all graph-based models. The models are trained using the Adam optimizer with a learning rate of 0.0005 for 40 epochs. As proof of stabilization we ran our model as well as the baselines with seeds 42, 123, 3407, 2025, 9999. Layer normalization and dropout are used to improve training stability and reduce overfitting during graph aggregation..

Due to the highly imbalanced nature of fraud detection datasets, evaluation is performed using PR-AUC, ROC-AUC, Precision, Recall, and F1-score across both LODO and OOD experimental settings. In addition, Expected Calibration Error (ECE) and Brier Score are used to evaluate calibration performance and prediction reliability under cross-domain distribution shifts. To better handle class imbalance and domain variability, a dynamic threshold is em-

ployed based on validation performance rather than using a fixed classification threshold.

4) Implementation

We implement our DG-Hetero-GNN and baseline models using PyTorch 2.3, PyTorch Geometric 2.7 and Python 3.11. For the DG-Hetero-GNN and its baselines we use the sourcecode¹.

All experiments are implemented using PyTorch and PyTorch Geometric on Google Colab with NVIDIA Tesla T4 GPU acceleration.

¹<https://github.com/raayanahmed-ai/DomainGeneralizedHeteroGNN>

5) Evaluation Metrics

Since there are imbalance nature of datasets, we evaluate the models performance using F1-score, Precision, Recall, and ROC-AUC.

F1-score: The F1-score measures the balance between precision and recall, making it suitable for imbalanced classification tasks. A higher F1-score shows a good performance in identifying fraud patterns.

Precision: Precision measures the proportion of correctly identified positive occurrences among all the predicted positives. It shows the model's ability to minimize false positives. Higher Precision indicates fewer false alarms.

Recall: Recall measures the proportion of actual positive instances that are correctly identified by the model. It reflects the model's ability to detect fraud transactions. Higher Recall represents better fraud coverage.

PR-AUC: The Area Under the Precision-Recall Curve (PR-AUC) evaluates the trade-off between precision and recall across different thresholds. PR-AUC is important for highly imbalanced fraud detection datasets as it focuses on the minority fraud class. Higher PR-AUC values show improved fraud detection capability.

ROC-AUC: The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) evaluates the model's ability to distinguish between fraudulent and legitimate samples across different classification thresholds. A higher ROC-AUC indicates a better ranking performance.

ECE: Expected Calibration Error (ECE) measures how well predicted probabilities align with

actual outcomes. Lower ECE values indicate better calibrated confidence estimates under cross-domain distribution shifts.

Brier Score: The Brier Score evaluates the accuracy of probabilistic predictions by measuring the mean squared error between predicted probabilities and true labels. Lower Brier Scores indicate more reliable probability estimation.

Additionally, we use a confusion matrix which includes True positives (TP), True Negatives (TN), False Positives (FP), False Negatives (FN), to provide a detailed understanding of model performance under various evaluation settings.

B. OVERALL EVALUATION

1) Statistics of Constructed Heterogeneous Financial Graphs

Table 1 shows the statistics of the heterogeneous financial graphs constructed from the Elliptic, IEEE-CIS, DGraphFin, and AMLSim datasets. These datasets contain different variations in the number of transaction nodes, entity nodes, and graph edges, which demonstrates strong distribution variations across domains. IEEE-CIS contains the largest number of transaction nodes and edges, while DGraphFin includes a large number of entity nodes representing complex relational interactions.

In contrast to this, AMLSim is shown as a smaller unseen target domain to evaluate the performance of the proposed framework under limited-data OOD conditions. As shown in Table 1, AMLSim contains significantly fewer nodes and edges compared to the other datasets. This is because AMLSim is a synthetic dataset which has an extremely low fraud ratio. This results in severe class imbalance. Under such conditions, evaluation metrics such as PR-AUC and F1-score may become unstable and less informative for cross-domain fraud evaluation.

Therefore, a controlled subset of AMLSim was constructed to preserve and maintain a meaningful fraud-risk ratio while maintaining heterogeneous transaction relationships and diverse fraudulent behavior patterns. This enables more reliable analysis of fraud detection performance under realistic OOD settings while preserving the structural diversity required for cross-domain evaluation.

2) LODO Cross-Domain Evaluation

Table 2 and 3a- 3c summarizes the Leave-One-Domain-Out (LODO) evaluation results. DG-Hetero-GNN achieves a strong ROC-AUC, PR-

AUC, and F1-score values across multiple unseen domains, demonstrating improved cross-domain fraud detection capability. In the LODO₀ setting, the proposed framework achieves a larger PR-AUC and F1-score. Although GAT achieves slightly higher ROC-AUC values and XGBOOST with improved PR-AUC in some LODO settings, DG-Hetero-GNN maintains a balanced performance across all evaluation metrics.

The LODO evaluation results show that homogeneous graph models such as GCN, GraphSAGE, and GAT achieves a competitive performance across different domains. However, these models are limited in clearly modeling heterogeneous financial interactions across multiple node and edge types. Similarly, feature-based methods such as MLP and XGBoost perform considerably well in several LODO settings, but they do not directly capture relational graph dependencies. HeteroGraphSAGE improves heterogeneous representation learning through heterogeneous neighborhood aggregation, yet DG-Hetero-GNN achieves more balanced performance across ROC-AUC, PR-AUC, and F1-score metrics under cross-domain evaluation. These results highlight the importance of combining heterogeneous graph modeling with domain generalization mechanisms for robust fraud detection under unseen domain shifts.

3) Detailed LODO Analysis

Table 3 presents a detailed analysis of DG-Hetero-GNN under different LODO settings using the metrics such as Precision, Recall, Threshold, ECE, Brier Score, and confusion matrix metrics. The results show that fraud detection under cross-domain conditions remains a challenge due to severe class imbalance and distribution shifts. The model helps maintain a relatively high recall values across all domains, indicating a strong capability in identifying fraudulent transactions.

The calibration analysis further demonstrates the reliability of our framework under unseen domains. Lower ECE and Brier scores indicate improved probability calibration and prediction confidence. The dynamically selected thresholds vary across domains, showing that different financial datasets require different decision boundaries for optimal fraud detection. However in some settings the ECE and Brier is moderate. But in fraud detection calibration of fraud detection is hard and analyzing metrics as a whole is better. This validates the importance

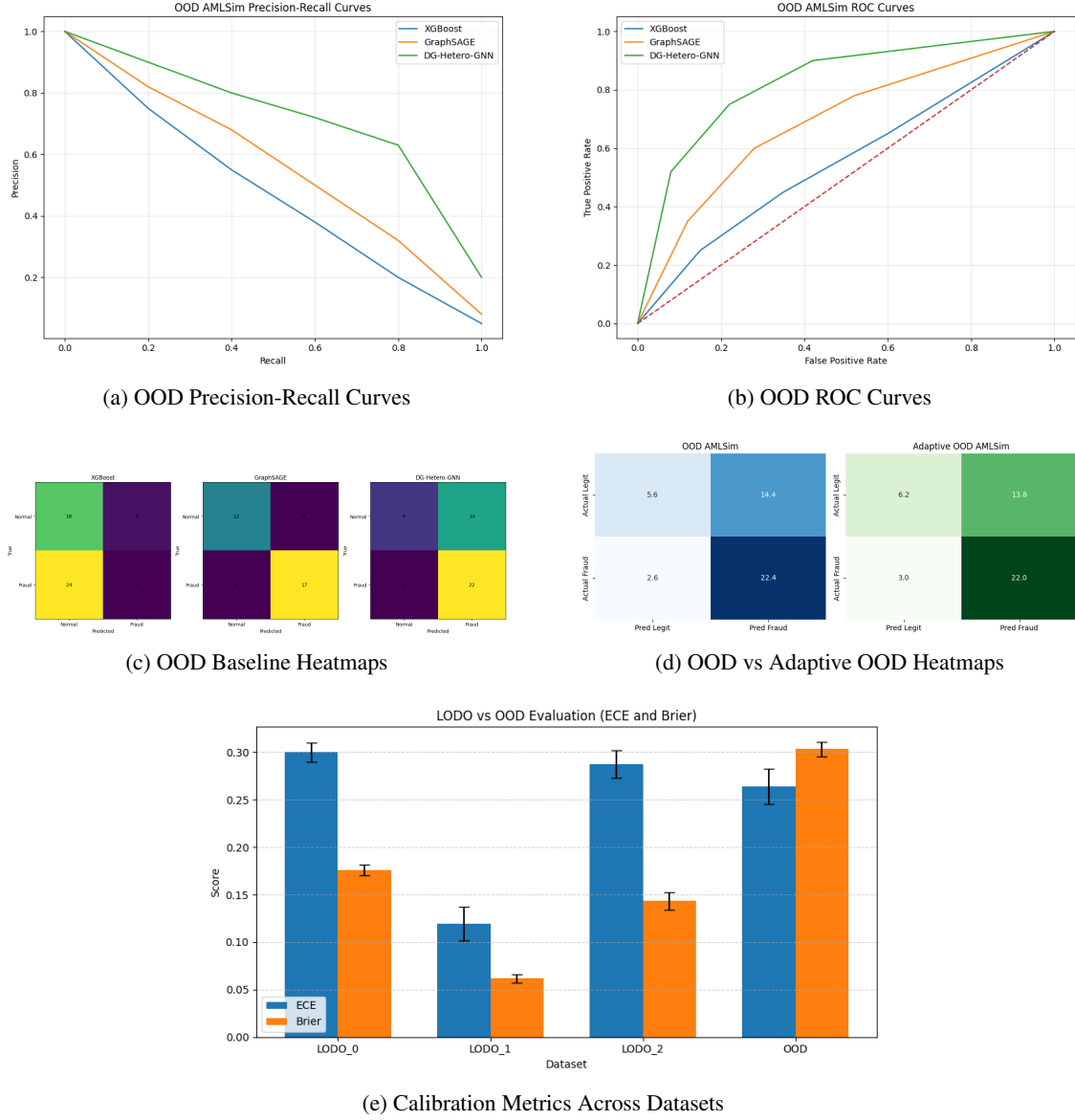


FIGURE 2: OOD evaluation and calibration analysis of DG-Hetero-GNN

of adaptive threshold optimization in cross-domain fraud detection systems.

4) OOD AMLSim Evaluation

Table 4 shows the Out-of-Distribution (OOD) evaluation using AMLSim as the unseen target domain. DG-Hetero-GNN achieves the highest ROC-AUC and F1-score among all baseline models, demonstrating improved fraud classification capability under realistic unseen-domain conditions. The PR-AUC results further show that the proposed model maintains strong precision-recall performance despite severe imbalance in fraud labels.

Among the baseline methods, GraphSAGE and GAT show competitive results due to their neighborhood aggregation mechanisms, while GCN and MLP show a lower performance under domain shifts. XGBoost achieves a strong calibration performance with lower ECE and Brier scores. However, its fraud classification performance remains weaker because tree-based methods cannot fully capture graph relational dependencies. HeteroGraphSAGE improves heterogeneous representation learning but lacks the domain generalization capability integrated within DG-Hetero-GNN.

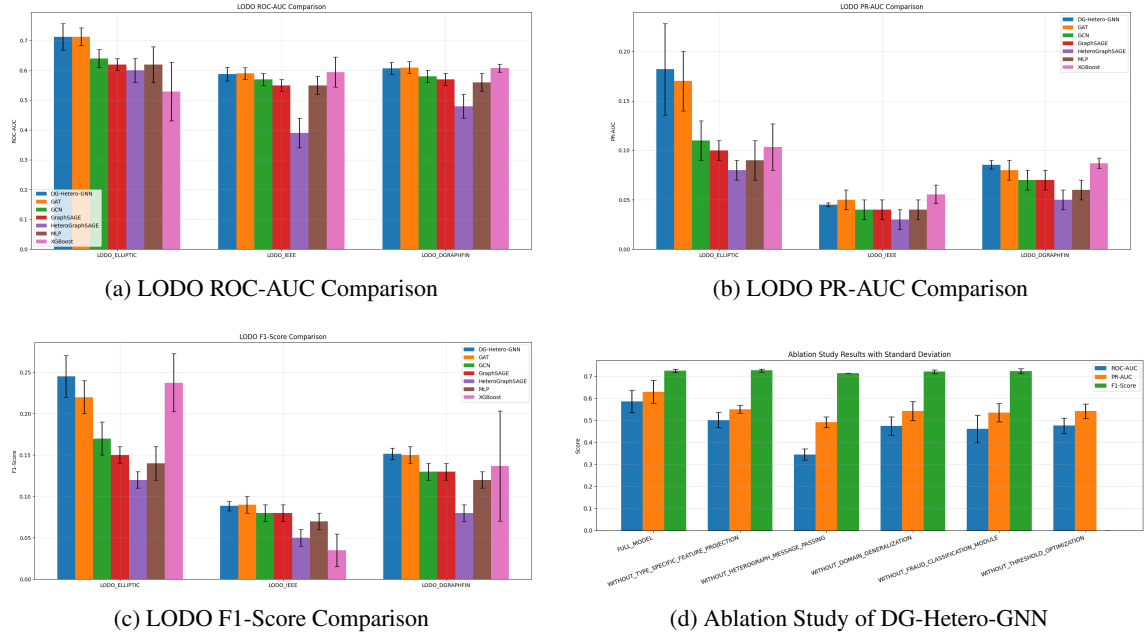


FIGURE 3: LODO evaluation and ablation analysis of DG-Hetero-GNN

5) Adaptive OOD Evaluation

Table 5 shows a comparative study between the standard OOD setting with the adaptive threshold setting. The adaptive approach maintains a competitive ROC-AUC, Precision, Recall, and F1-score values while improving calibration quality through lower ECE and Brier scores. The adaptive setting also improves True Negative predictions while reducing calibration mismatch under unseen target-domain conditions.

These results demonstrate that adaptive threshold optimization helps align the decision boundary with the shifted probability distribution of the unseen domain. This improves prediction reliability without requiring access to target-domain labels. This makes the approach suitable for real-world fraud detection applications where distribution shifts frequently occur.

6) ROC and Precision-Recall Curve Analysis

Fig. 2(a) and Fig. 2(b) present Precision-Recall and ROC curves under OOD AMLSim evaluation. DG-Hetero-GNN consistently achieves a larger PR and ROC areas compared with baseline approaches, showing a stronger discrimination capability between fraud and safe transactions.

The Precision-Recall curves show the effective-

ness of the proposed model under severe class imbalance. While XGBoost and GraphSAGE experience rapid precision degradation at higher recall values, DG-Hetero-GNN maintains a relatively stable precision across different recall levels. Similarly, the ROC curves demonstrate that DG-Hetero-GNN achieves stronger true positive rates at lower false positive rates under unseen-domain settings.

7) Heatmap and Calibration Analysis

Fig. 2(c) and Fig. 2(d) visualize the confusion matrix heatmaps for baseline and adaptive OOD settings. DG-Hetero-GNN shows improved fraud detection capability by maintaining higher True Positive predictions compared to these baseline methods. The adaptive threshold setting further improves prediction balance by reducing false classifications under a shifted target-domain distributions.

Fig. 2(e) presents the calibration analysis across datasets using ECE and Brier Score metrics. The results show that calibration performance varies across domains due to differences in transaction distributions and fraud characteristics. The adaptive threshold setting improves prediction confidence alignment, demonstrating the importance of calibration-aware fraud detection under OOD environments.

8) Ablation Study

Fig. 3(d) presents the ablation study of DG-Hetero-GNN under OOD. Removing type-specific feature projection, heterogeneous graph message passing, domain generalization, deep fraud classification, or threshold optimization leads to noticeable performance degradation across ROC-AUC, PR-AUC, and F1-score metrics. In particular, removing heterogeneous graph message passing causes the largest drop in ROC-AUC and PR-AUC. This highlights the importance of modeling heterogeneous financial interactions between transactions, accounts, and merchants.

In addition to this, removing domain generalization reduces cross-domain robustness under OOD settings, while replacing the deep fraud classification module with a shallow classifier decreases representation learning capacity. The threshold optimization component is also critical under highly imbalanced fraud detection settings. When threshold optimization is removed, the model uses a fixed decision threshold of 0.5 for binary classification. Due to severe class imbalance and distribution shift across domains, predicted fraud probabilities are often skewed toward lower values. This causes the model to classify most transactions as legitimate. Consequently, the F1-score collapses despite relatively stable ROC-AUC and PR-AUC values. These findings show that each proposed component contributes meaningfully towards robust and generalized fraud detection performance across heterogeneous financial domains.

C. DISCUSSION

The experimental results show that DG-Hetero-GNN improves fraud detection under cross-domain and out-of-distribution environments. The combination of heterogeneous graph learning, adaptive threshold optimization, calibration-aware evaluation, and domain generalization helps the framework to achieve a strong stability across multiple financial datasets making it effective in real world fraud detection.

VI. LIMITATIONS

Even though DG-Hetero-GNN shows a stable cross-domain fraud detection performance and improves generalization across heterogeneous financial datasets, there are still certain areas that can be improved. First, the current framework operates on static graph representations and does not model temporal transaction dynamics. While the heterogeneous graph structure effectively captures rela-

tional dependencies between entities, using temporal graph modeling can improve the detection of evolving fraud patterns. Second, the integration of type-specific feature projection, heterogeneous message passing, domain generalization, calibration, and adaptive threshold optimization uses additional computational complexity for large-scale transaction networks. However, these components also help our framework achieve robustness and cross-domain adaptability. Third, although the framework achieves strong predictive performance, integrating explainability mechanisms could further improve transparency and trustworthiness for practical financial decision-making environments. In addition, evaluation on a broader real-world and streaming financial systems may provide further insights into long-term scalability and deployment feasibility.

VII. FUTURE WORK

In the future, improving adaptability, scalability, and real-world deployment capability of the DG-Hetero-GNN could be an important direction. One possible extension is incorporating temporal and dynamic graph modeling to capture evolving fraudulent transaction behaviors effectively. Improving computational efficiency for industrial-scale financial transaction networks could also be explored through lightweight graph learning strategies and optimized training mechanisms. In addition to this, integrating explainable artificial intelligence techniques may help provide more transparent and interpretable fraud predictions for practical financial applications. The framework can also be evaluated on more diverse multi-institutional and streaming financial datasets to further strengthen its robustness, adaptability, and generalization across evolving fraud scenarios.

VIII. CONCLUSION

In this paper, we propose DG-Hetero-GNN for cross-domain financial fraud detection under heterogeneous and OOD settings. This framework combines heterogeneous GraphSAGE aggregation, domain generalization, focal loss, and adaptive threshold optimization to improve fraud detection robustness under distribution shifts. Experimental results on Elliptic, IEEE-CIS, DGraphFin, and AMLSim demonstrate competitive ROC-AUC, PR-AUC, and F1-score performance under LODO and OOD evaluation settings. Overall, DG-Hetero-GNN presents as an effective framework for robust fraud detection across heterogeneous financial domains.

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